Ecogeographic diversity in maxillary palate morphology: early explorations with a new database

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Recent research suggests that cross-population disparities in vocal tract morphology may impact the development of languages by yielding subtle articulatory biases. Relatedly, findings in biological anthropology indicate that ecogeographic factors yield cross-population disparities in the upper vocal tract, specifically regarding nasal cavity morphology. This variation also impacts the maxillary palate in ways that may yield minor influences on articulatory biases. Here we present exploratory findings from a new database, which contains CT scans of maxillary palates from six individuals from six populations of distinctive ecologies. We demonstrate how CT scans can be utilized to measure maxillary palate characteristics. These measurements can be analyzed in ways that we illustrate here in a pilot study for six individuals in the sample. We suggest that this database could elucidate ecogeographic influences on cross-population differences in vocal tract characteristics. Clearer understanding of such differences could, in turn, better inform potential hypotheses on cross-population effects on articulatory biases.

1. Introduction

Studies in biological anthropology offer evidence of cross-population variation in upper vocal tracts, due in part to environmental characteristics. Most notably, this variation has been observed in studies of nasal cavities. In especially dry and cold climates, such as the Arctic Circle, indigenous individuals have been found to exhibit narrower and taller nasal cavities; thought to aid in the humidification of inspired air while maintaining adequate oxygen intake (Kelly et al., 2023). Though the nasal and oral cavities are distinct anatomical structures, they are interconnected components of the human respiratory system and may be shaped by similar evolutionary pressures. Further, the nasal and oral cavities, along with the pharynx, comprise the vocal tract. Population variation within these cavities

is relevant to our understanding of potential cross-population differences in articulatory biases. Recent genetic evidence supports this connection, showing associations between maxillary morphology, nasal cavity morphology, and climatic variables (Evteev & Movsesian, 2016). Due to the floor of the nasal cavity and the hard palate sharing a common foundation within the maxilla, it follows that climatic adaptation influencing the nasal cavity may also affect the hard palate. This paper describes the approach of a new pilot project investigating how geographic and climatic factors shape hard palate morphology and, in turn, potentially impact phonetic patterns via variable articulatory biases.

The latter point, i.e. the potential ramifications of such variation on articulatory biases, relates to a growing body of research contending that an "anatomical bias" towards certain sound types exist across groups. Blasi et al. (2019) observed that populations with overbite/overjet configurations are more likely to rely on labiodental sounds. In a similar vein, Moisik and Dediu (2017) studied hard palate morphology in populations speaking "click" languages in southern and eastern Africa. They found that variation in palate shape creates biases that influence the ease of click production. Producing click sounds is facilitated by a smooth, concave palate with a central gap that allows the tongue to generate a pressure gradient. Their analysis of palatal arch and contour between the canines, premolars, and molars revealed that Bushmen populations generally exhibit narrower, shallower palates lacking a pronounced alveolar ridge, while Himba individuals have "shelved" palates with differing anterior and posterior depths. They also observed that a prominent alveolar ridge increases articulatory effort and muscle force, even though tongue-to-palate distance is reduced. Further, other population morphological differences have been related to palatal concavity (Fuchs et al., 2005). Additionally, other research suggests that palatal characteristics may impact articulatory biases in a way which has long-term effects on the development of individual languages, by creating biases in the transmission of vowel types (Dediu, Janssen and Moisik 2019).

2. Methods and sample

We assembled a dataset of approximately 400 cranial computed tomography (CT) scans representing diverse human populations worldwide. These include samples from six populations occupying distinct ecological contexts: Zulu, Southern Europe, Equatorial Africa, Arctic, East Asia, and Australia. The majority of crania analyzed date to the 19th and 20th centuries, with the exception of the Arctic sample, which derives from individuals dated to approximately the 5th–17th centuries and collected from present-day Alaska. For this preliminary analysis, a

single cranial CT scan was selected from each population. By sampling across markedly different climatic regimes, this illustrative subset is intended to provide a conceptual framework for future, larger-scale assessments of climatic influences on hard palate morphology. All specimens included in this study were accessed and analyzed with permission from their respective repositories and curating institutions.

CT data were reconstructed into three-dimensional models and analyzed using 3D Slicer (version 5.3.0; Fedorov et al., 2012), an open-source medical imaging platform. Each model was aligned to the Frankfurt Horizontal Plane prior to measurement to maintain anatomical consistency. A total of seven linear measurements were recorded, including anteroposterior (A–P) length, mediolateral (M–L) breadth, and palate vault angle. In addition, two spline-based curvature measurements were taken to quantify three-dimensional curvature of the hard palate and capture variation in palatal contour beyond traditional linear dimensions (Figure 1). All landmark placements and measurements were performed by the same observer and repeated to evaluate intra-observer error. Reliability was assessed using the technical error of measurement (TEM) and Bland–Altman plots, both of which indicated high reproducibility and minimal measurement bias. This protocol provides a standardized and repeatable approach to quantifying hard palate morphology, facilitating comparisons across geographically and climatically diverse populations and contributing to our understanding of environmentally driven craniofacial adaptation.

3. Results

Comparative analysis of hard palate morphology revealed clear differences in palatal dimensions and curvature (Table 1). The individuals from Australia, South Africa, and the Arctic exhibited the longest anteroposterior palates, whereas individuals from Equatorial Africa, Europe, and East Asia showed comparatively shorter palates (Figure 2). In contrast, mediolateral breadth was greatest among the South African, Australian, and Equatorial African individuals, while the European, East Asian, and especially Arctic specimens demonstrated narrower palates. Variation in palatal curvature was also evident, with the East Asian, European, and Australian specimens displaying relatively irregular or undulating surfaces, while the Equatorial African and South African individuals exhibited smoother and more uniform palatal contours. Differences in palatal vault height further distinguished these individuals: the South African, Arctic, and Equatorial African specimens presented the tallest vaults, whereas the European, East Asian, and Australian samples possessed shallower vaults.

Overall, the largest pairwise differences across individuals were observed between the Australian group and most other groups, across measurements (Table 1). Preliminary analyses comparing overall facial size to palate size using geometric mean (geomean) scaling are forthcoming and are expected to clarify the role of size-related allometry in shaping palate morphology across climatic regions.

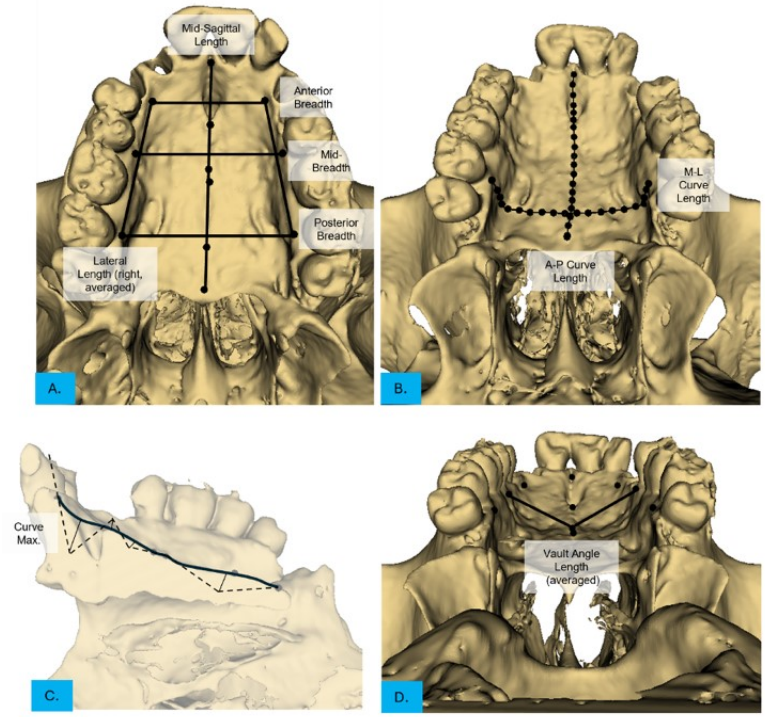

Figure 1. Depiction of the measurements collected from each sample.

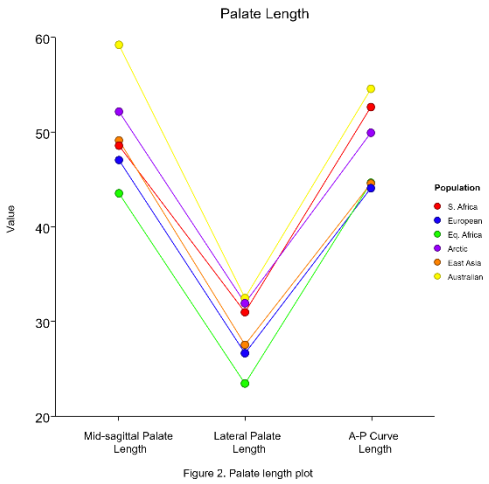

Figure 2. Measurements of three key variables (Mid-sagittal palate length, lateral palate length, and A-P curve length) for the six individuals.

Variable	Population Comparison	Highest differences
Palate vault angle	Australia – S. Europe	4.100
Lateral palate length	Australia – Eq. Africa	9.049
Anteroposterior curve length	Australia – Eq. Africa	15.660
Anteroposterior curve mean	Australia – Eq. Africa	0.106
Anteroposterior curve maximum	Arctic – Eq. Africa	0.233
Mediolateral curve length	Australia – S. Europe	15.620
Mediolateral curve mean	Eq. Africa – East Asia	-0.129
Mediolateral curve maximum	Eq. Africa – East Asia	-0.782
Mid-sagittal palate length	Australia – S. Europe	10.503
Anterior maxillary breadth	Australia – S. Europe	7.814
Mid-maxillary breadth	Australia – S. Europe	11.761
Posterior maxillary breadth	S. Europe - S Africa	-12.061

Table 1. Illustrative comparative analysis of variation across six individuals from distinct populations.

4. Discussion and conclusion

These illustrative results from six individuals are largely consistent with nasal morphology research. The nasal cavity and the hard palate share an anatomical location; the maxilla is both the floor of the nose and the roof of the mouth. As such, individuals ancestral to the Arctic possess the longest nasal passages while simultaneously possessing extremely narrow nasal floors (Kelly et al., 2023). Conversely, equatorial populations possess extremely broad, though not necessarily short nasal floors. These patterns are consistent in the hard palate. Long anteroposterior nasal palates are functionally advantageous for respiration in populations with larger average body sizes (and high basal metabolic rates). Longer nasal passages theoretically enable a larger amount of overall oxygen intake and greater contact with nasal mucosa. However, equatorial Africans (hot-adapted) and Arctic (cold-adapted) individuals do not show the greatest opposition across all the measurement values we reported here. Indeed, there was a great number of opposing measurements between the Australian and Southern European groups (Table 1). Both findings contrast with expectations based on nasal morphology patterning, where such traits would more typically be anticipated in the Arctic sample.

Of course, we make no claims about population-level effects here given that we only report findings from one individual from each population. We are simply illustrating the approach we will be utilizing to more comprehensively analyze this database of CT scans, with an ultimate aim of elucidating cross-population variation in vocal tract morphology. Nevertheless, it is worth noting that the individuals in this sample inhabiting colder or drier climates, such as those from the Arctic and South Africa, tend to exhibit longer and taller palatal vaults, whereas individuals from warmer or more humid regions, including Equatorial Africa and East Asia, display shorter and broader palates with smoother curvature. These results are consistent with our current understanding of ecogeographic effects on vocal tract morphology (Kelly et al. 2023). These morphological differences may therefore reflect functional adaptations for regulating airflow in cold and dry air, features that could secondarily influence the range or ease of sound production. For example, a narrower anterior palate could place pressures related to the production of anterior obstruents. These pressures could include a stronger bias against multiple places of articulation for anterior fricatives within the same language.

While the limited sample size constrains the generalizability of these results, we offer the findings as an illustration of the method that could be applied to our large sample, yielding more quantitatively driven analyses on population-level

variation. Future work incorporating our larger dataset, advanced geometric morphometric techniques, and linguistic comparisons will be essential to shed light on the potential interaction between climate, craniofacial anatomy, and articulatory biases.

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